

DZALINDA, Russian Federation

WMO: 219080

Lat: 70.133N

Lon: 113.967E

Elev: 62

StdP: 100.58

Time Zone: 9.00 (E09)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-52.5	-50.3	-53.5	0.0	-52.2	-51.4	0.0	-50.2	7.4	-14.1	7.0	-19.5	0.7	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.9	17.3	24.3	16.1	21.8	14.8	18.4	24.7	16.9	22.9	15.4	20.8	2.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.3	21.4	14.3	10.2	19.3	12.8	9.3	17.9	52.5	24.7	47.8	23.4	43.5	20.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.4	5.3	-53.8	30.6	2.2	2.1	-55.4	32.2	-56.7	33.4	-57.9	34.6	-59.5	36.1		
			-53.8	22.0	2.2	3.2	-55.4	24.3	-56.7	26.1	-57.9	27.9	-59.5	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.4	-34.6	-35.5	-23.3	-10.8	-1.0	11.7	14.1	10.4	2.3	-10.0	-26.9	-34.9
	DBStd	19.69	9.84	9.25	9.71	7.07	6.66	5.95	4.26	3.90	4.27	6.86	8.90	8.58
	HDD10.0	8111	1382	1275	1032	623	344	51	11	42	232	621	1108	1392
	HDD18.3	10879	1640	1508	1290	873	598	212	145	247	480	879	1358	1650
	CDD10.0	300	0	0	0	0	5	101	137	55	2	0	0	0
	CDD18.3	27	0	0	0	0	0	13	13	1	0	0	0	0
	CDH23.3	306	0	0	0	0	1	136	141	28	0	0	0	0
	CDH26.7	71	0	0	0	0	0	36	31	4	0	0	0	0
(14) Wind	WSAvg	1.9	1.8	1.5	1.7	2.3	2.3	2.5	2.0	1.9	1.8	1.7	1.7	1.5
(15) Precipitation	PrecAvg	265	8	7	10	12	23	37	47	50	27	22	15	9
	PrecMax	396	20	16	20	31	54	63	152	102	70	44	34	20
	PrecMin	187	1	1	2	1	3	10	18	11	7	8	4	1
	PrecStd	51	5	4	5	8	12	14	28	25	15	9	8	5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.2	-7.7	1.3	10.4	21.4	30.1	29.9	26.4	19.4	4.7	-1.9	-8.8
		MCWB	-7.9	-8.2	-1.0	5.5	12.7	19.4	17.5	18.0	13.1	3.2	-2.9	-9.3
	2%	DB	-12.8	-14.4	-1.2	6.3	16.2	27.2	27.2	23.3	13.5	2.3	-7.0	-14.3
		MCWB	-13.2	-14.7	-3.0	2.9	9.7	16.8	17.7	15.8	8.7	1.4	-7.5	-14.7
	5%	DB	-15.7	-18.1	-4.0	3.4	12.3	24.3	24.8	20.8	10.9	0.6	-10.9	-18.4
		MCWB	-16.0	-18.4	-5.2	0.7	7.1	15.7	16.5	14.9	7.4	-0.2	-11.5	-18.7
	10%	DB	-19.4	-22.2	-7.5	0.8	9.3	21.5	22.1	18.3	8.7	-1.2	-14.5	-22.4
		MCWB	-19.7	-22.5	-8.5	-1.3	5.1	14.1	15.2	13.5	6.3	-2.0	-14.8	-22.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.1	-8.0	-0.7	5.5	13.2	20.2	20.6	19.7	13.8	3.6	-2.4	-9.3
		MCDB	-7.3	-7.3	1.0	9.9	21.1	26.3	26.9	25.3	18.7	4.4	-1.6	-8.7
	2%	WB	-13.2	-14.7	-3.1	2.7	9.7	18.0	18.6	17.0	9.6	1.5	-7.5	-14.5
		MCDB	-12.9	-14.4	-1.3	5.7	15.4	25.7	24.9	21.4	12.4	2.0	-7.0	-14.1
	5%	WB	-16.0	-18.4	-5.4	1.0	7.4	16.4	17.2	15.3	7.8	-0.2	-11.4	-18.7
		MCDB	-15.6	-18.1	-4.0	3.2	12.1	23.2	23.3	19.5	10.2	0.6	-10.9	-18.4
	10%	WB	-19.6	-22.4	-8.5	-1.4	5.2	14.6	15.8	13.8	6.5	-1.9	-14.8	-22.6
		MCDB	-19.3	-22.2	-7.5	0.7	9.0	20.9	21.0	17.7	8.7	-1.2	-14.5	-22.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.1	13.9	13.6	9.5	9.5	9.3	10.1	7.7	6.7	7.8	7.8
		MCDBR	12.6	10.9	11.2	14.8	12.3	13.3	13.5	15.7	13.3	6.2	9.2	10.8
		MCWBR	12.2	10.7	9.9	12.1	8.1	6.8	6.7	9.9	9.0	5.4	9.0	10.6
	5% WB	MCDBR	12.8	11.0	10.9	14.1	11.9	12.8	12.3	13.4	11.1	5.9	9.3	10.9
		MCWBR	12.5	10.8	9.7	11.8	8.1	7.3	6.9	9.5	8.0	5.3	9.1	10.6
(40) Clear-Sky Solar Irradiance	taub		N/A	0.215	0.284	0.313	0.341	0.352	0.411	0.394	0.306	0.224	N/A	N/A
	taud		N/A	1.992	2.042	2.077	2.165	2.427	2.265	2.264	2.442	2.074	N/A	N/A
	Ebn at Noon		0	501	697	805	830	840	759	718	711	475	0	0
	Edh at Noon		0	40	83	112	117	95	108	96	59	35	0	0
(44) All-Sky Solar Radiation	RadAvg		0.00	0.41	1.94	4.20	5.68	5.55	4.90	3.28	1.62	0.51	0.00	0.00
	RadStd		0.00	0.03	0.10	0.21	0.36	0.48	0.33	0.39	0.17	0.05	0.00	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page